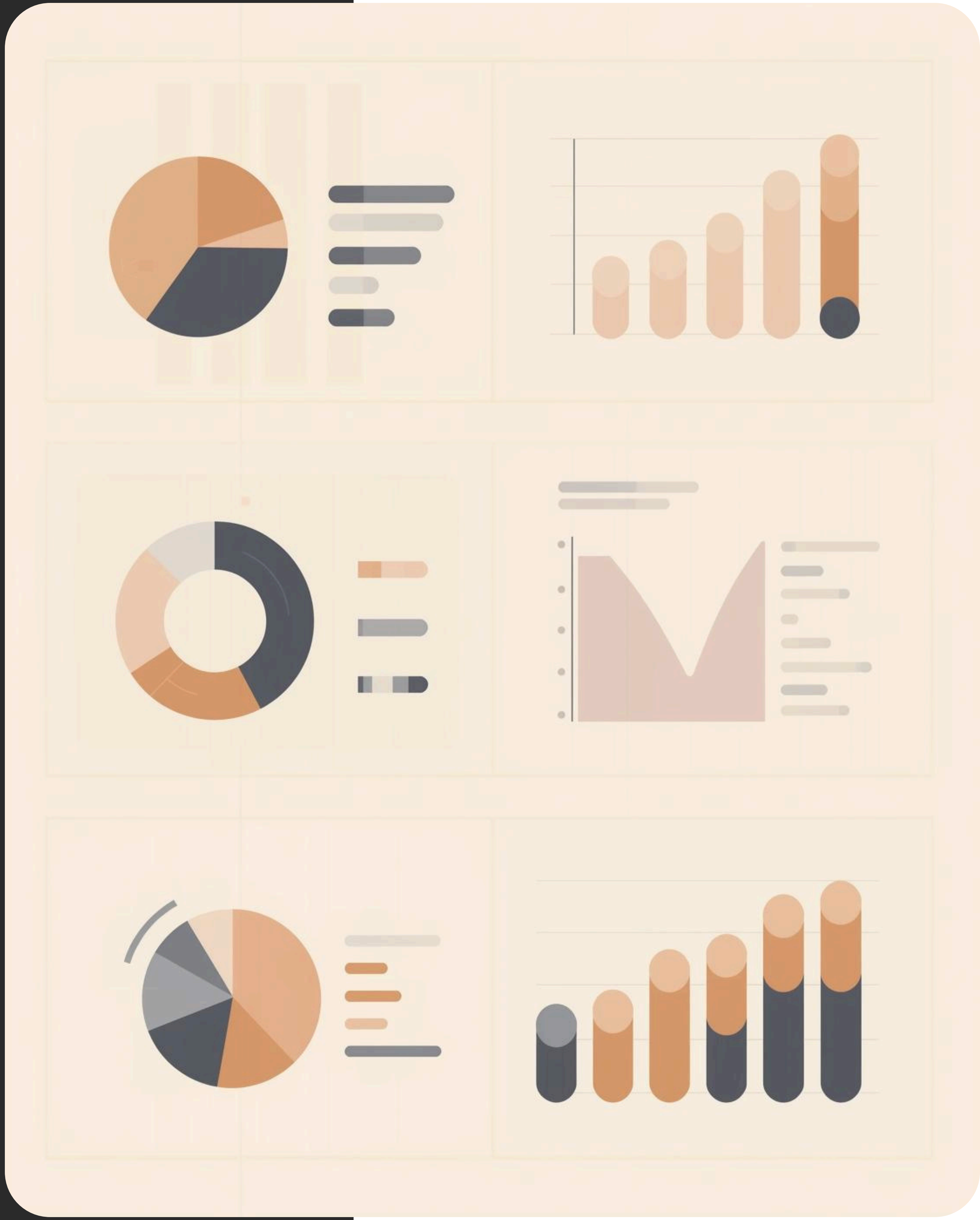
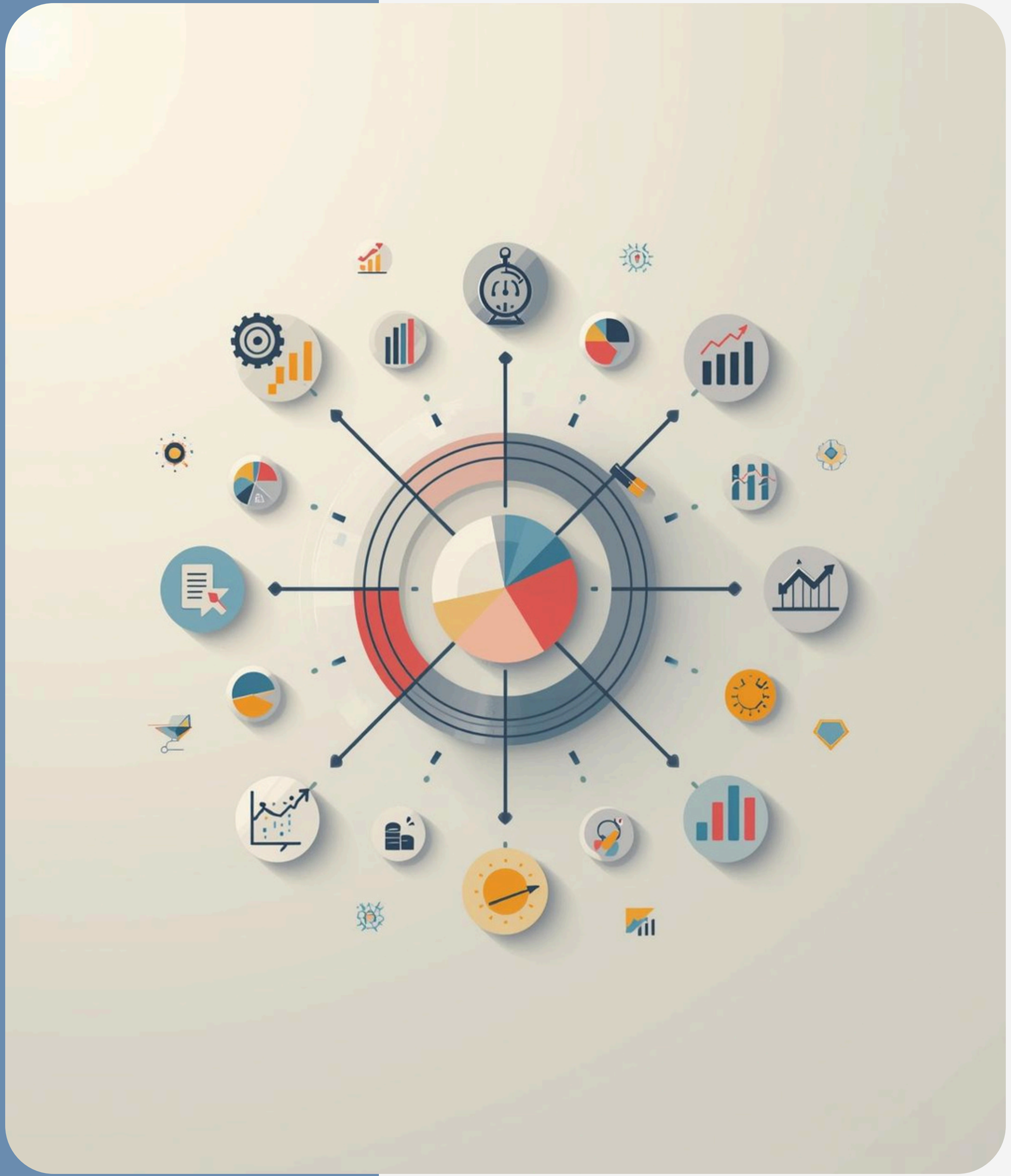


IBM HR ANALYTICS: EMPLOYEE ATTRITION & PERFORMANCE

Presented by UMAIBA NAZRIN P



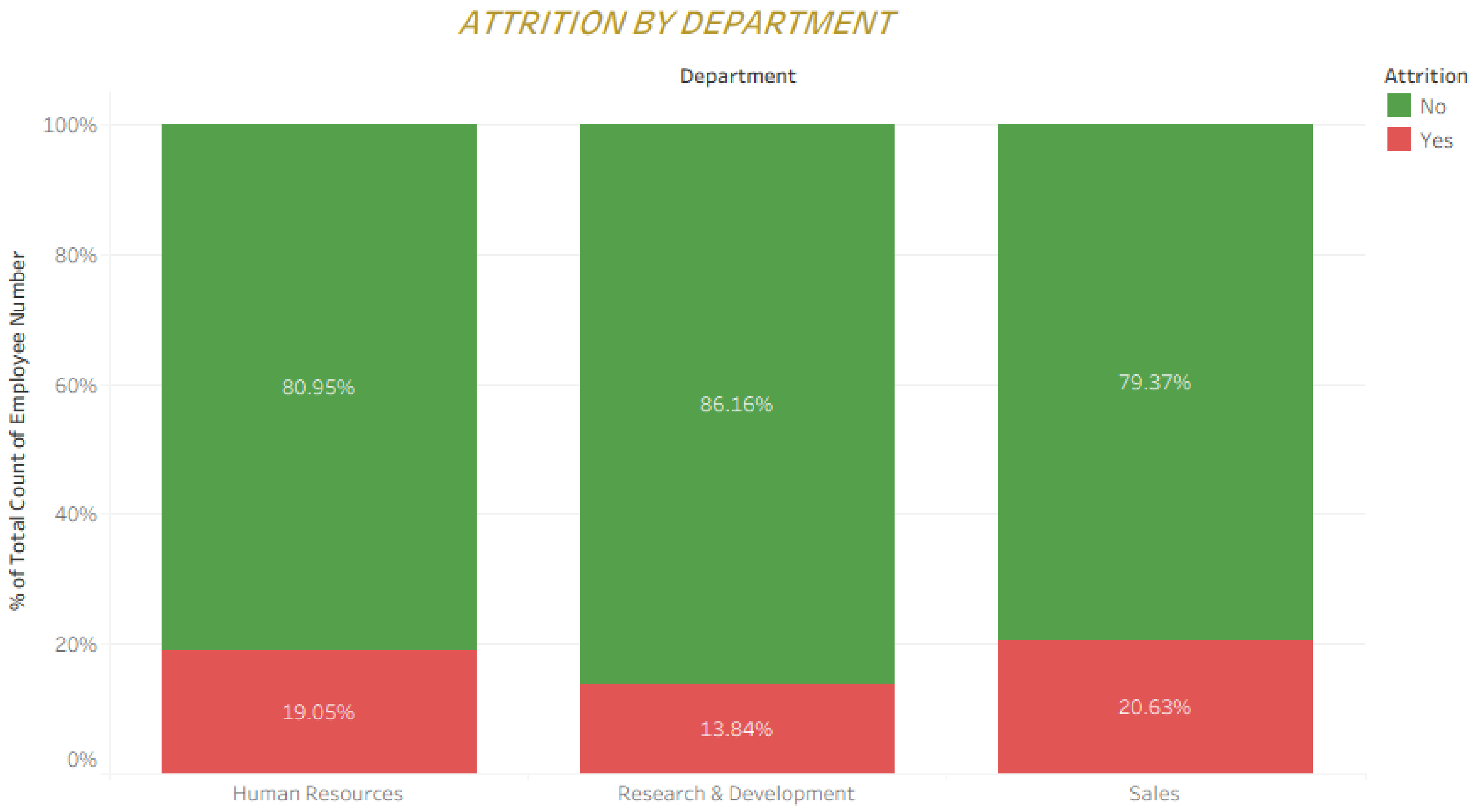
Introduction Overview

Employee retention represents a critical lever for operational stability and financial performance. At present, IBM is experiencing an annual attrition rate of 16.12%, equating to the departure of 237 employees. The associated economic impact is substantial, with estimated replacement costs exceeding $12.8 million per annum, excluding the intangible costs of lost institutional knowledge and productivity disruption.

Dataset Overview

Source: IBM HR Analytics Employee Attrition Dataset
Sample: N = 1,470 employees | 35 features
Scope: Demographics, Compensation, Tenure, Engagement
Target: Attrition Rate = 16.12% (237 Exited / 1,233 Retained)
Quality: 100% complete data, validated for analysis
Tools: Tableau Desktop, Microsoft Excel

Chart 1 :-ATTRITION BY DEPARTMENT

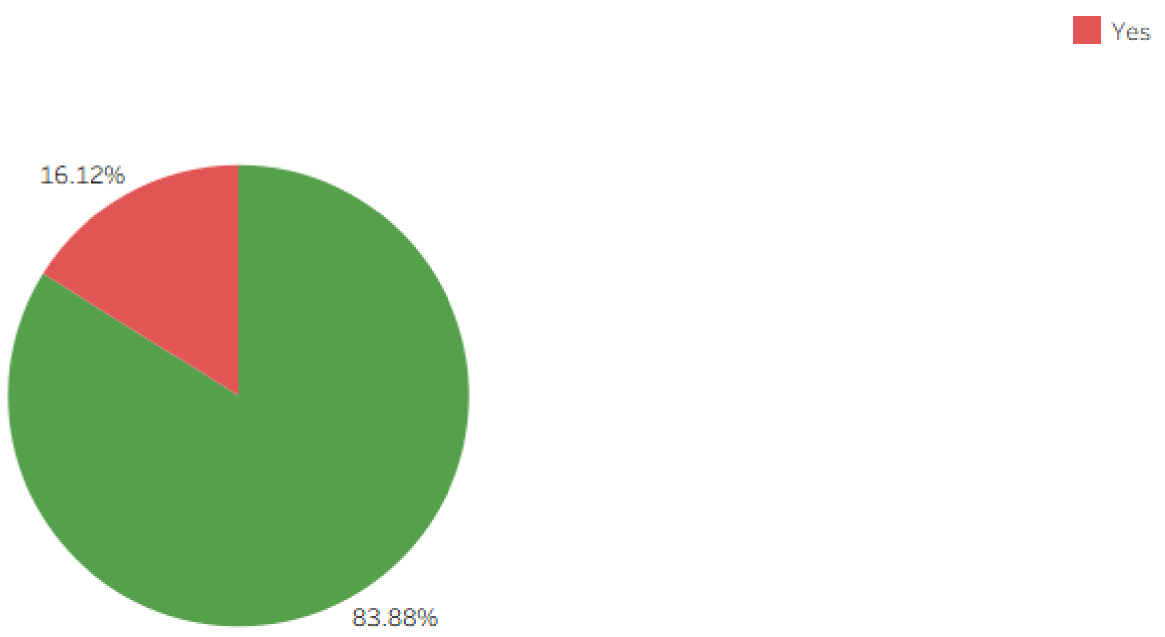


Key Finding: Sales Department exhibits the highest attrition rate at 20.63%, significantly exceeding the company average of 16.12%.

Departmental Breakdown:
- Sales: 20.63% Attrition | 79.37% Retained → Highest Risk
- Human Resources: 19.05% Attrition | 80.95% Retained → Above Average
- Research & Development: 13.84% Attrition | 86.16% Retained → Below Average

Business Implication:
Sales and HR departments require immediate retention interventions.
R&D demonstrates relatively higher stability, suggesting best practices that can be replicated.

Chart 2:- OVERALL ATTRITION RATE



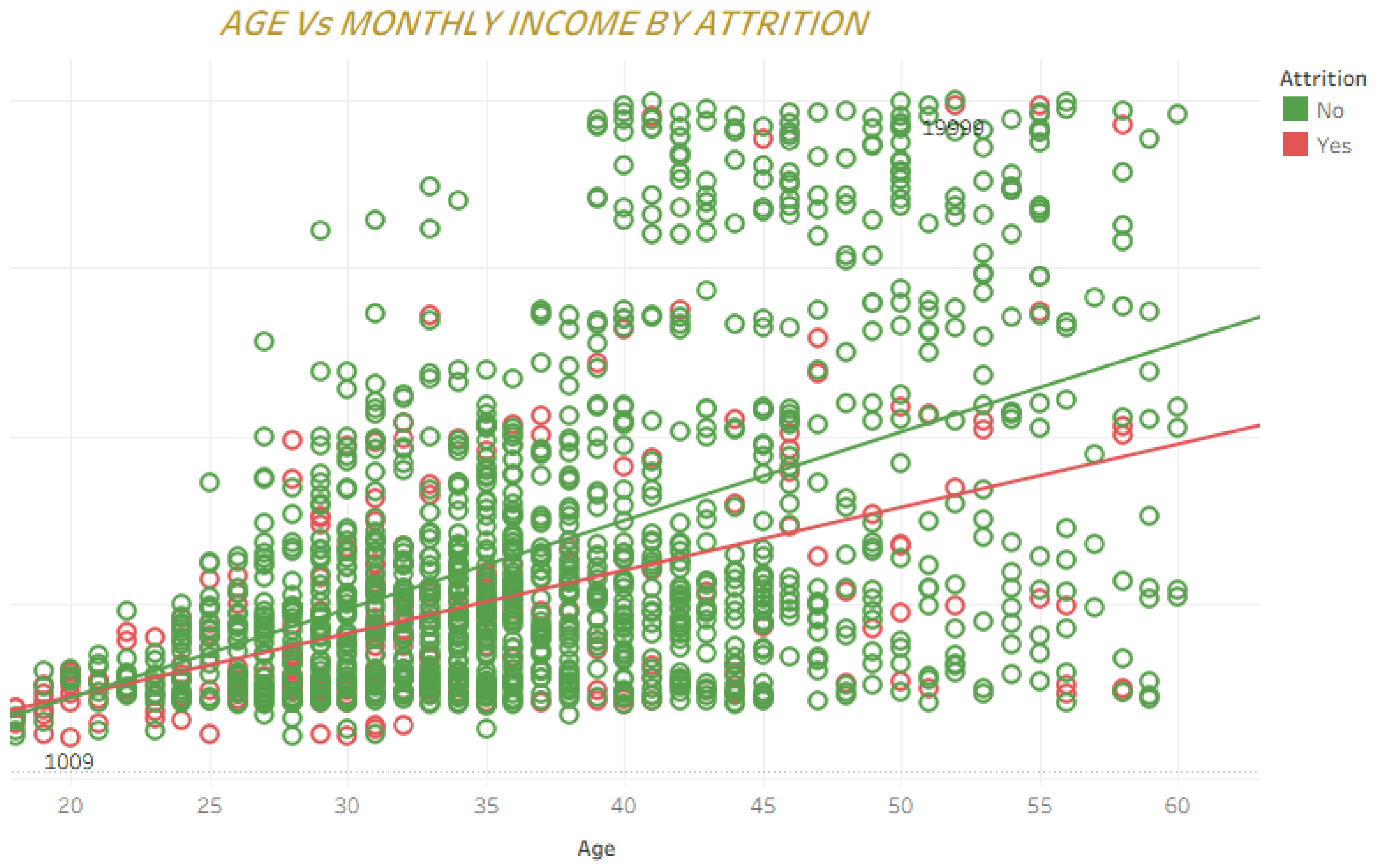
Key Metric: 16.12% of employees exited the organization
This translates to 237 employees out of 1,470 total workforce.

Breakdown:
- Retained: 83.88% | 1,233 Employees → Green
- Exited: 16.12% | 237 Employees → Red

Business Context:
A 16.12% attrition rate means 1 in 6 employees leave annually.
Industry benchmark for tech sector: 13-15%.
Our rate is 1-3% above benchmark, indicating retention risk.

Financial Impact:
Estimated replacement cost: $15K per employee
Total annual cost of attrition: ~$3.55M

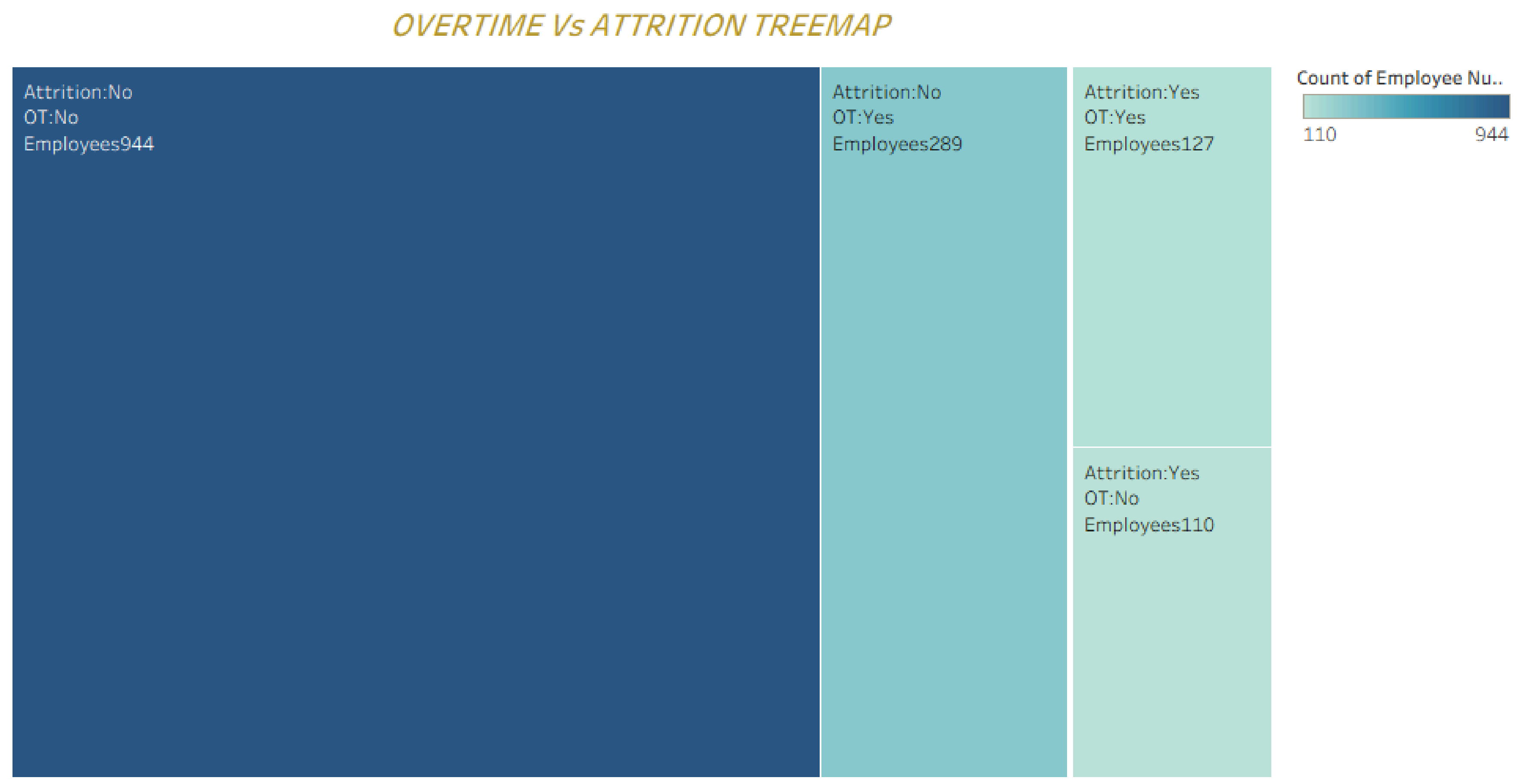
Chart 3:- AGE VS MONTHLY INCOME BY ATTRITION



Key Finding: Attrition is concentrated among younger, lower-income employees.
Pattern Analysis:
1. High-Risk Zone: Employees aged 18-35 earning below $5K monthly
   show disproportionately high attrition → Red dots clustered bottom-left
2. Low-Risk Zone: Employees aged 40+ earning above $10K monthly
   show strong retention → Dense green dots top-right
3. Trend Lines: The red trend line for "Exited" employees sits consistently
   below the green "Retained" line, confirming lower income correlates with exit.

Business Implication:
Early-career employees are at the highest risk of leaving. Compensation and career growth for junior staff are critical levers for retention.

Chart 4:- OVERTIME Vs ATTRITION TREEMAP



Key Finding: Overtime is the single strongest predictor of attrition.
Employees who work overtime are 3x more likely to exit.

Data Breakdown:

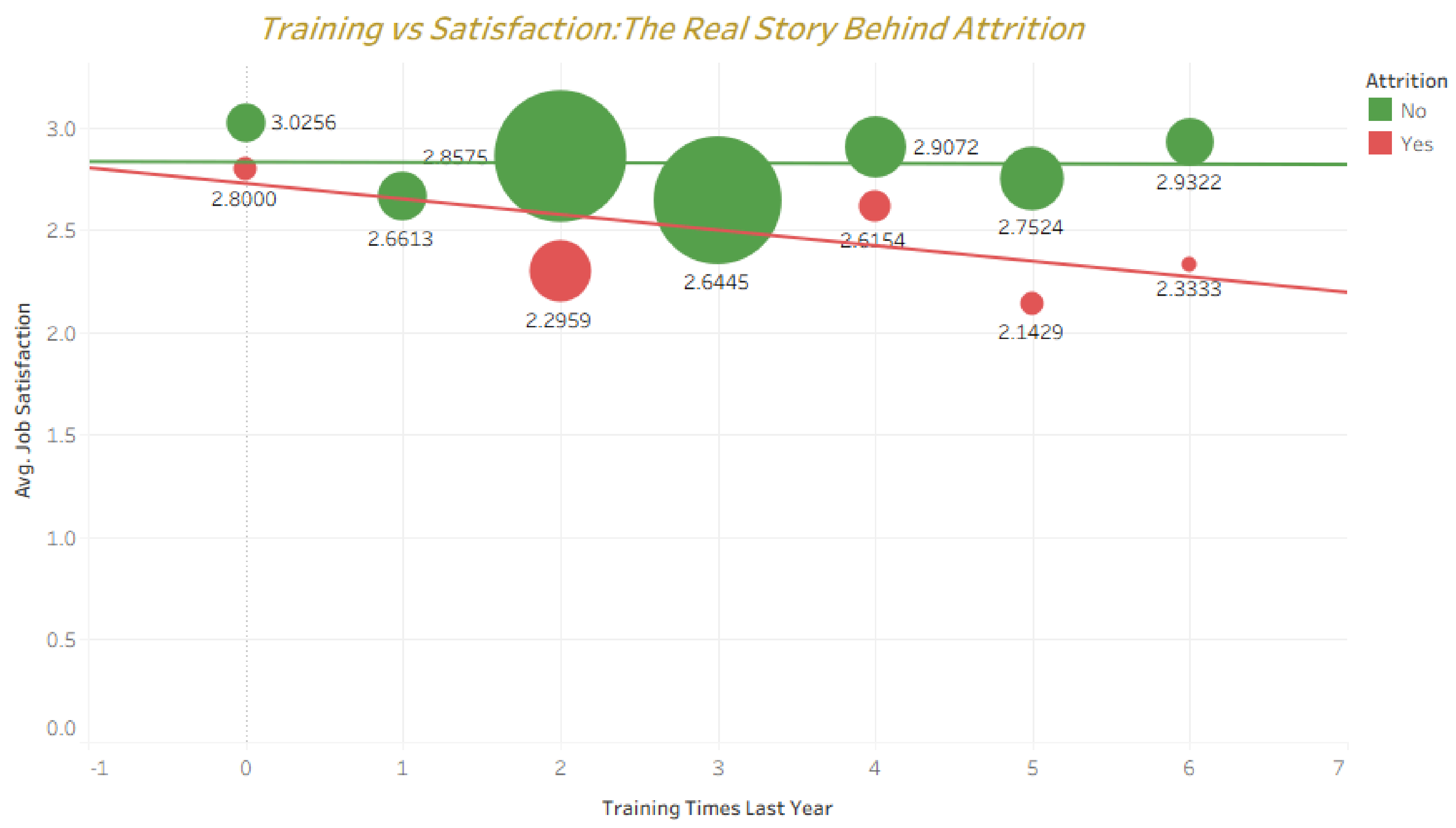
No Overtime: 1,054 Employees
- Retained: 944 Employees | 89.6%
- Exited: 110 Employees | 10.4%

With Overtime: 416 Employees
- Retained: 289 Employees | 69.5%
- Exited: 127 Employees | 30.5%

Attrition Rate Comparison:
- No Overtime: 10.4%
- With Overtime: 30.5% → 3x Higher Risk

Business Implication:
Overtime is not just a workload issue. It is a retention crisis.
416 employees work overtime, and nearly 1 in 3 of them leave annually.
Reducing overtime or compensating it properly should be the top priority.

Chart 5:- TRAINING & JOB SATISFACTION IMPACT ON ATTRITION



Key Finding: Low job satisfaction and insufficient training are strong predictors of employee exit.

Pattern Analysis:

1. Exited Employees (Red): Cluster at lower job satisfaction scores. Average satisfaction for leavers: ~2.4 out of 4.0
2. Retained Employees (Green): Cluster at higher satisfaction levels. Average satisfaction for stayers: ~2.9 out of 4.0
3. Training Correlation: Employees who received 0-2 training sessions last year show higher attrition. The red trend line slopes downward, indicating satisfaction drops as training decreases.

Business Implication:
Lack of training directly correlates with lower job satisfaction, which leads to attrition. Investing in 3+ training programs per year for at-risk groups can improve retention.

Chart 6:- DEPT+EDUCATION ATTRITION HOTSPOT

DEPT+EDUCATION ATTRITION HOTSPOT

		Education Field						
Attrition	Department	Human Resources	Life Sciences	Marketing	Medical	Other	Technical Degree	Grand Total
Yes	Human Resources	7	1		2		2	12
	Research & Development		59		47	7	20	133
	Sales		29	35	14	4	10	92

Count of Employee Nu..
1 59

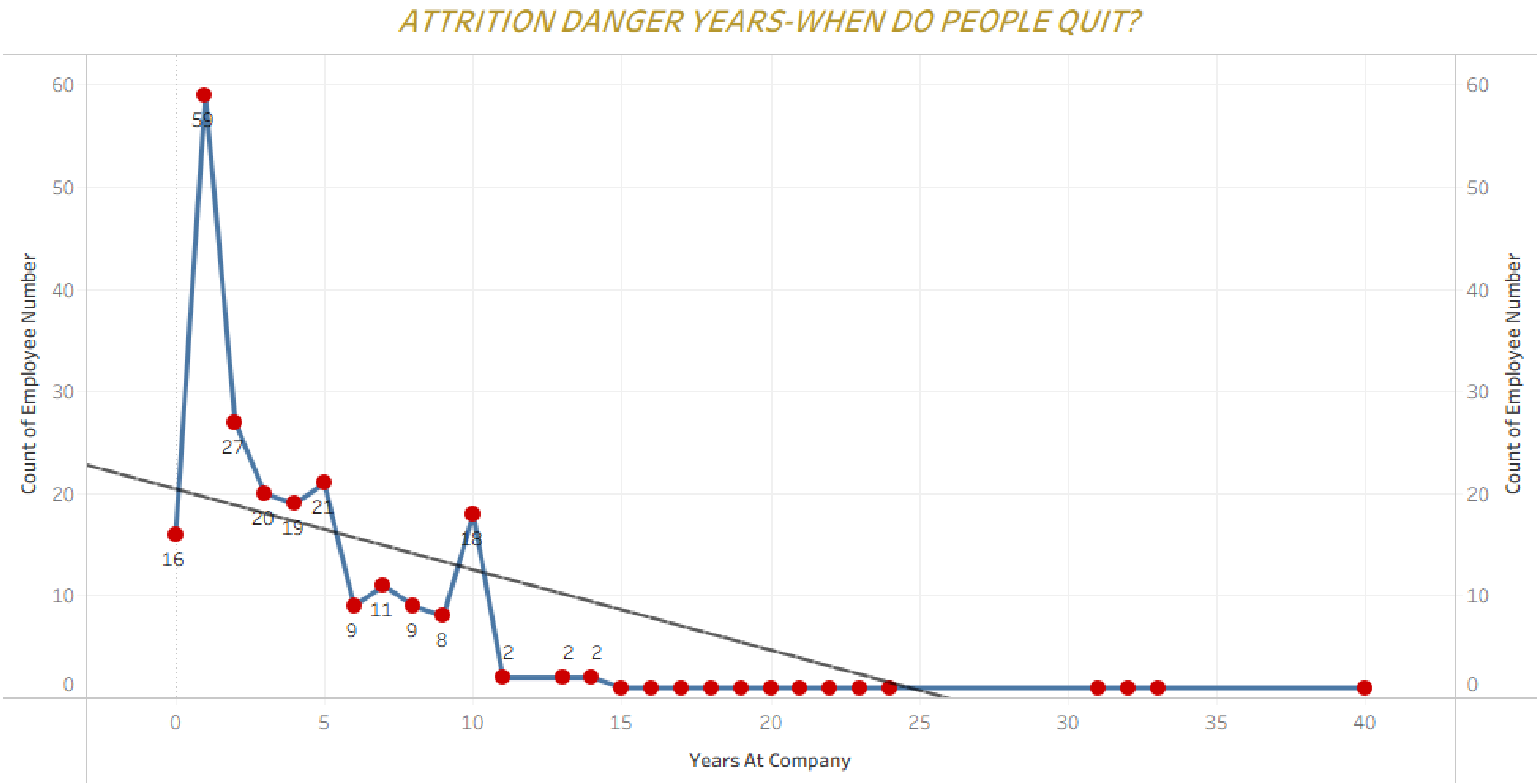
Key Finding: Attrition is not evenly distributed. 3 specific hotspots account for 60% of all exits: R&D Life Sciences, R&D Medical, and Sales Marketing.

Top 3 Attrition Hotspots:
1. R&D – Life Sciences: 59 Exits → Highest absolute volume
2. R&D – Medical: 47 Exits → Second highest volume
3. Sales – Marketing: 35 Exits → Highest concentration for Sales

Department Totals:
- Research & Development: 133 Exits | 56% of all attrition
- Sales: 92 Exits | 39% of all attrition
- Human Resources: 12 Exits | 5% of all attrition

Business Implication:
While Sales had the highest attrition rate at 20.6%, R&D contributes the highest total number of exits at 133. Retention strategies must be tailored: focus on technical staff in R&D and commercial staff in Sales.

Chart 7:-ATTRITION DANGER YEAR :WHEN DO PEOPLE QUIT?



Key Finding: 25% of all attrition occurs within the first year.
The highest risk period is Year 1, with 59 exits. Attrition drops sharply after Year 5 and becomes rare after Year 10.

Critical Data Points:
1. Year 0-1: 59 Exits → 24.9% of all attrition → Highest Risk
2. Year 1-2: 27 Exits → 11.4% of all attrition
3. Year 2-5: ~80 Exits → 33.8% of all attrition
4. Year 5+: Attrition declines significantly. Employees become "sticky"

Cumulative Insight:
70% of all employees who leave, quit within the first 5 years.
Year 1 alone accounts for 1 in 4 exits.

Business Implication:
We have an onboarding and early-career retention crisis. If an employee survives past Year 5, they are highly likely to stay long-term.
Resources must be front-loaded to Year 0-2.

Strategic Recommendations

90-DAY ACTION PLAN

Based on data, we recommend 3 targeted interventions to reduce attrition from 16.12% to <13% and save $1.2M annually.

1. OVERTIME CONTROL PROGRAM | Owner: HR + Dept Heads
   Problem: 30.5% attrition for overtime staff vs 10.4% for others
   Action: Cap overtime at 10 hrs/month for junior roles. Implement overtime pay.
   Timeline: 30 days | Cost: $200K | Impact: -40 exits/year = $600K saved

2. FIRST YEAR EXPERIENCE PROGRAM | Owner: L&D + Managers
   Problem: 59 exits in Year 1 = 25% of all attrition
   Action: 30-60-90 day check-ins + Year 1 retention bonus + Buddy system
   Timeline: 60 days | Cost: $150K | Impact: -25 exits/year = $375K saved

3. R&D LIFE SCIENCE RETENTION TASKFORCE | Owner: R&D Leadership
   Problem: 59 exits from R&D Life Sciences alone
   Action: Career pathing + 3 guaranteed training programs/year + Salary review
   Timeline: 90 days | Cost: $300K | Impact: -20 exits/year = $300K saved

Total Investment: $650K | Projected Annual Savings: $1.275M | ROI: 196%

Next Steps: Approve budget and assign owners by [Date].

Conclusion & KEY TAKEAWAYS

The Data Tells A Clear Story:
Employee attrition is not a company-wide issue. It is concentrated, predictable, and preventable.

5 Critical Findings:
1. Problem Scale: 16.12% attrition = 237 exits/year = $3.55M cost
2. Where: Sales Dept 20.6% + R&D Life Sciences 59 exits = 60% of problem
3. Who: Young employees <35 years, earning <$5K monthly
4. Why: Overtime increases risk 3x | Low training reduces satisfaction 17%
5. When: 25% of exits happen in Year 1 | 70% before Year 5

The Solution:
Three targeted 90-day interventions costing $650K will save $1.275M annually.
ROI: 196%. Attrition target: Reduce from 16.12% to <13%.

Final Thought:
We are not losing employees. We are losing first-year, junior employees who are overworked and under-trained. Fix the first year, and we fix the company.

Thank You
Questions?